# Filter Wizard

AAF Filter

Created on 05/26/2026

ANALOG DEVICES

# Filter Wizard Design Report

## Filter Requirements for Low-Pass, 4th order Butterworth

Specifications: Optimize: Power; +Vs: 5; -Vs: -5

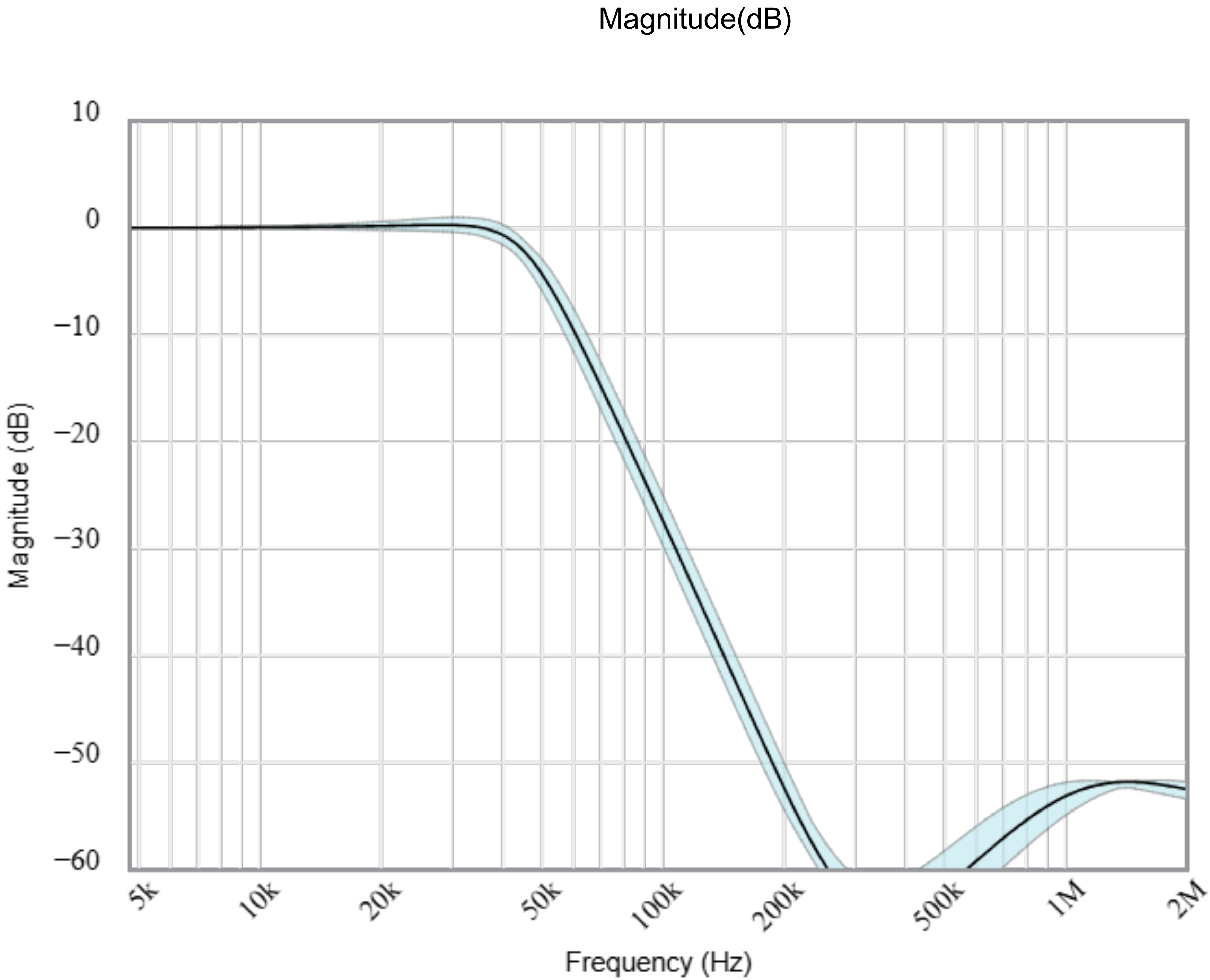
Gain: 0 dB

Passband: -3dB at 48kHz

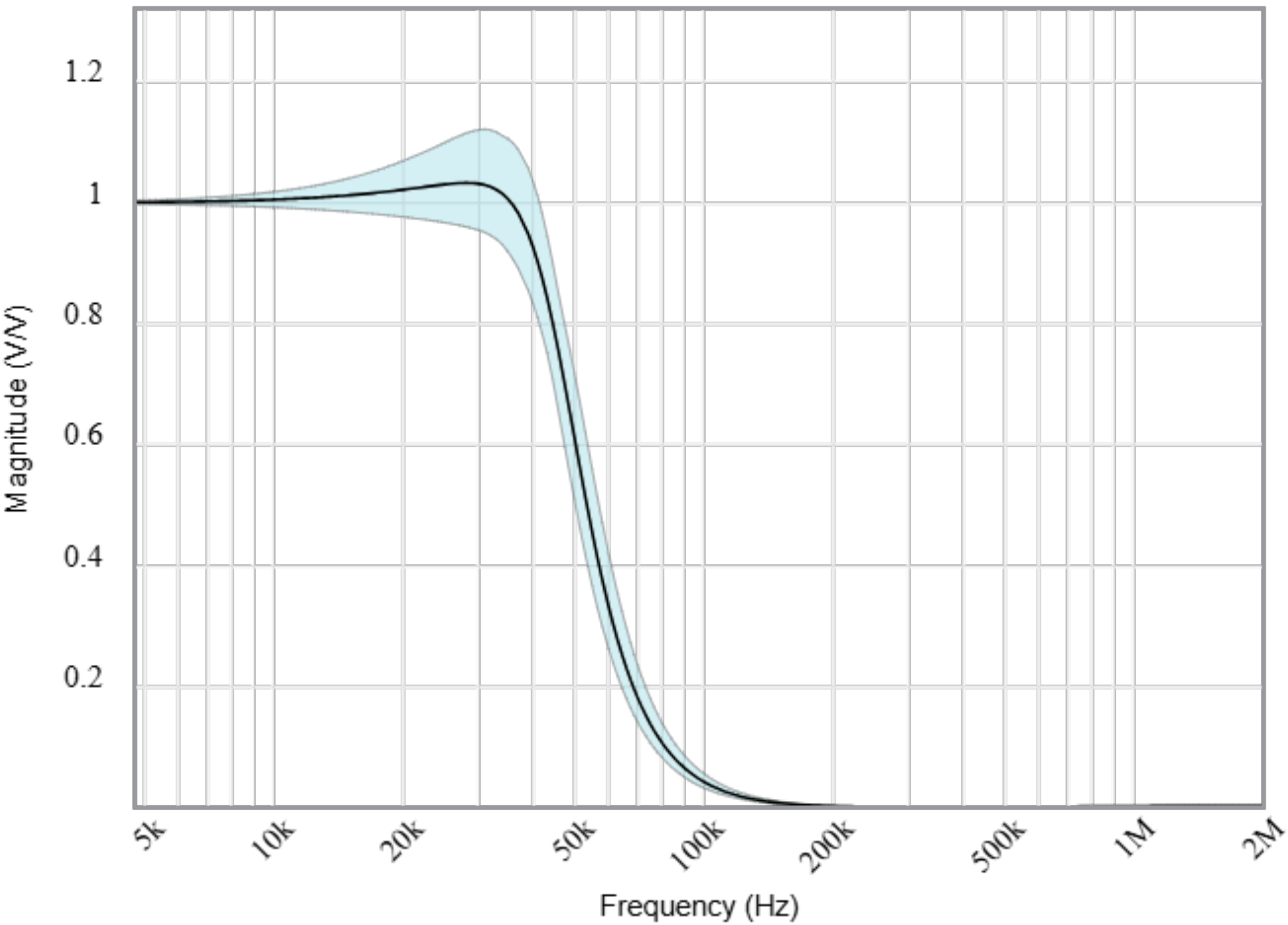
Stopband: -40dB at 200kHz

Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

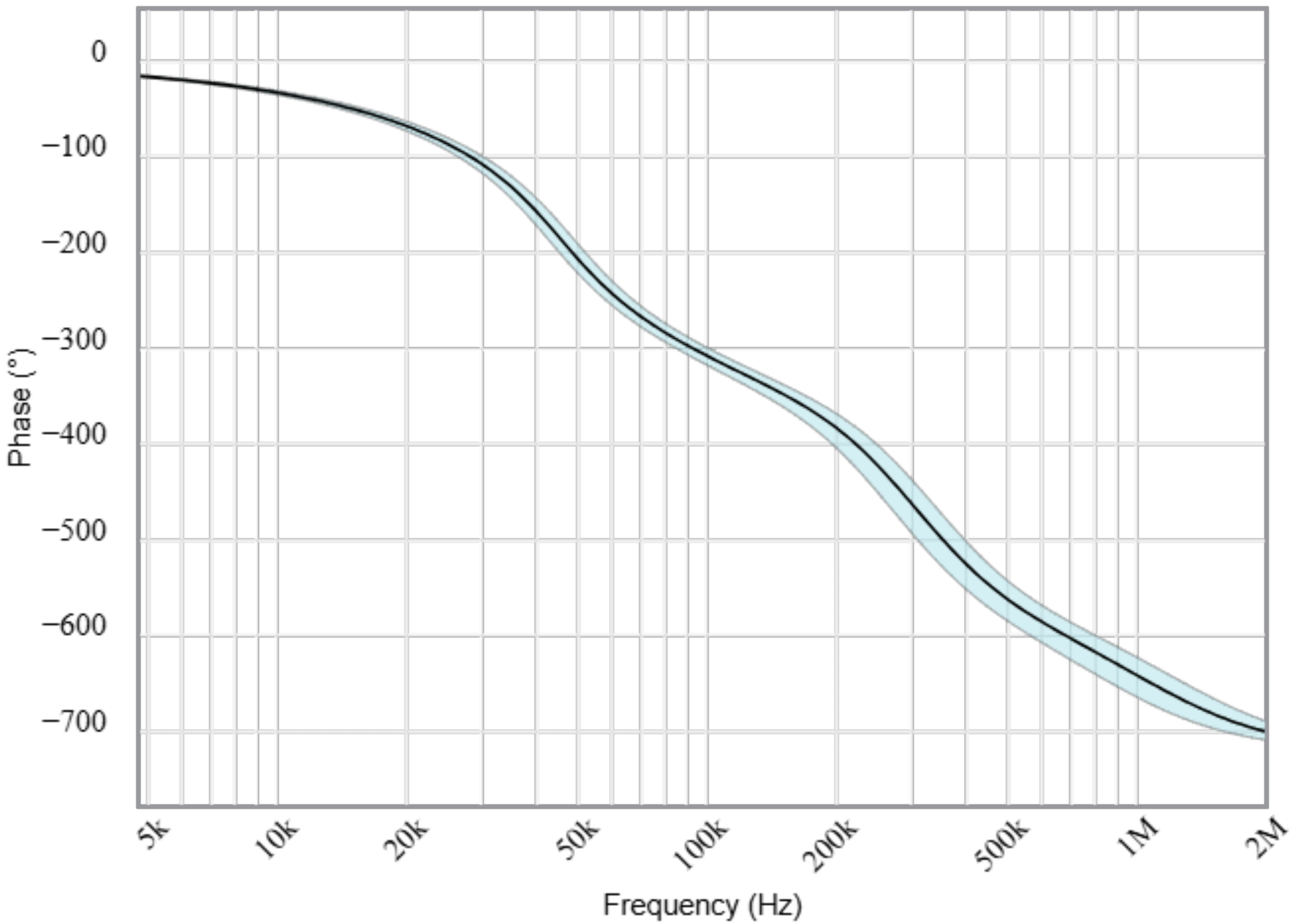
BOM: refer to BOM.csv file


Magnitude(dB)
10
0
−10
−20
−30
−40
−50
−60
Magnitude (dB)
5k
10k
20k
50k
100k
200k
500k
1M
2M
Frequency (Hz)

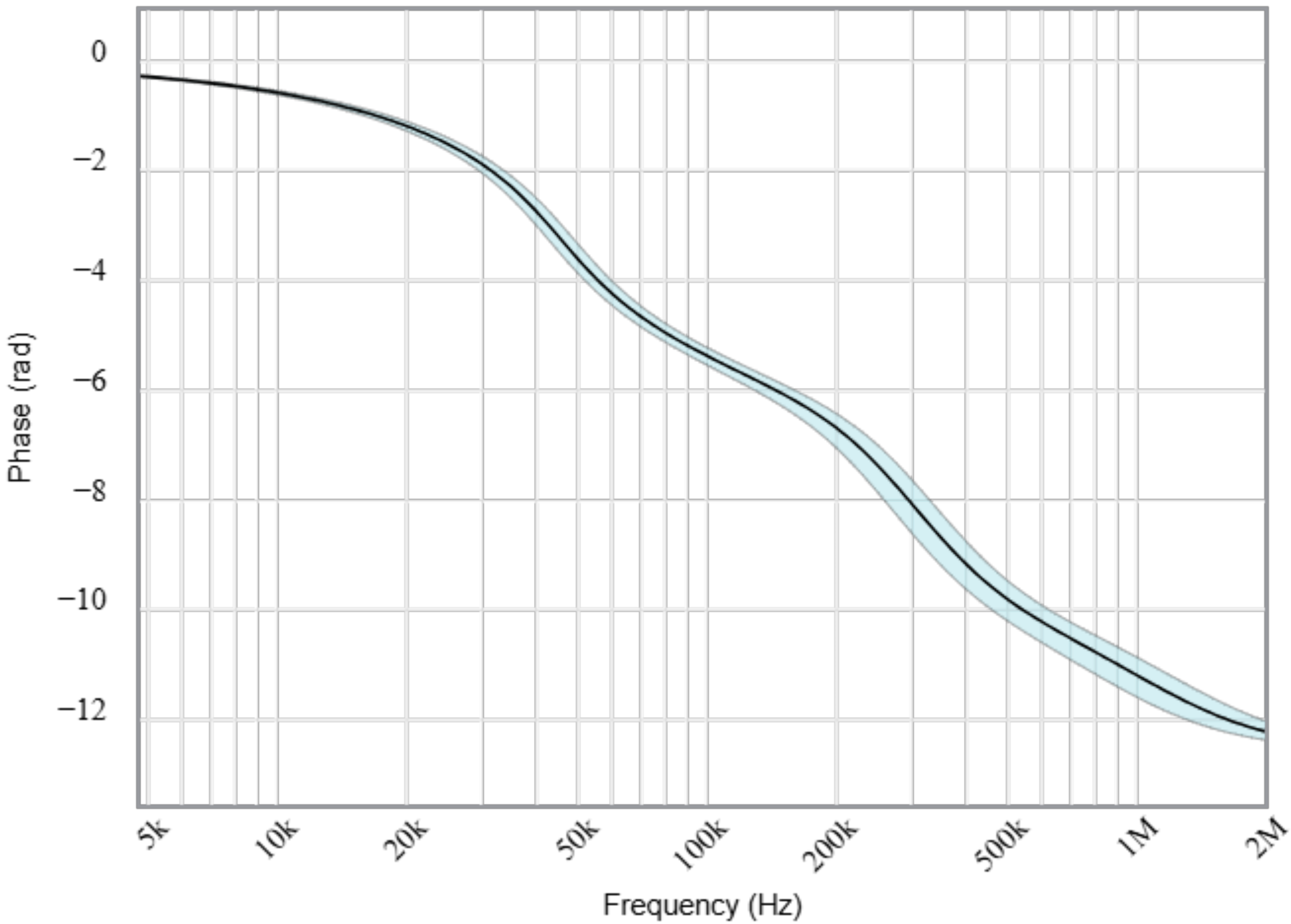
Magnitude(Volts per Volt)

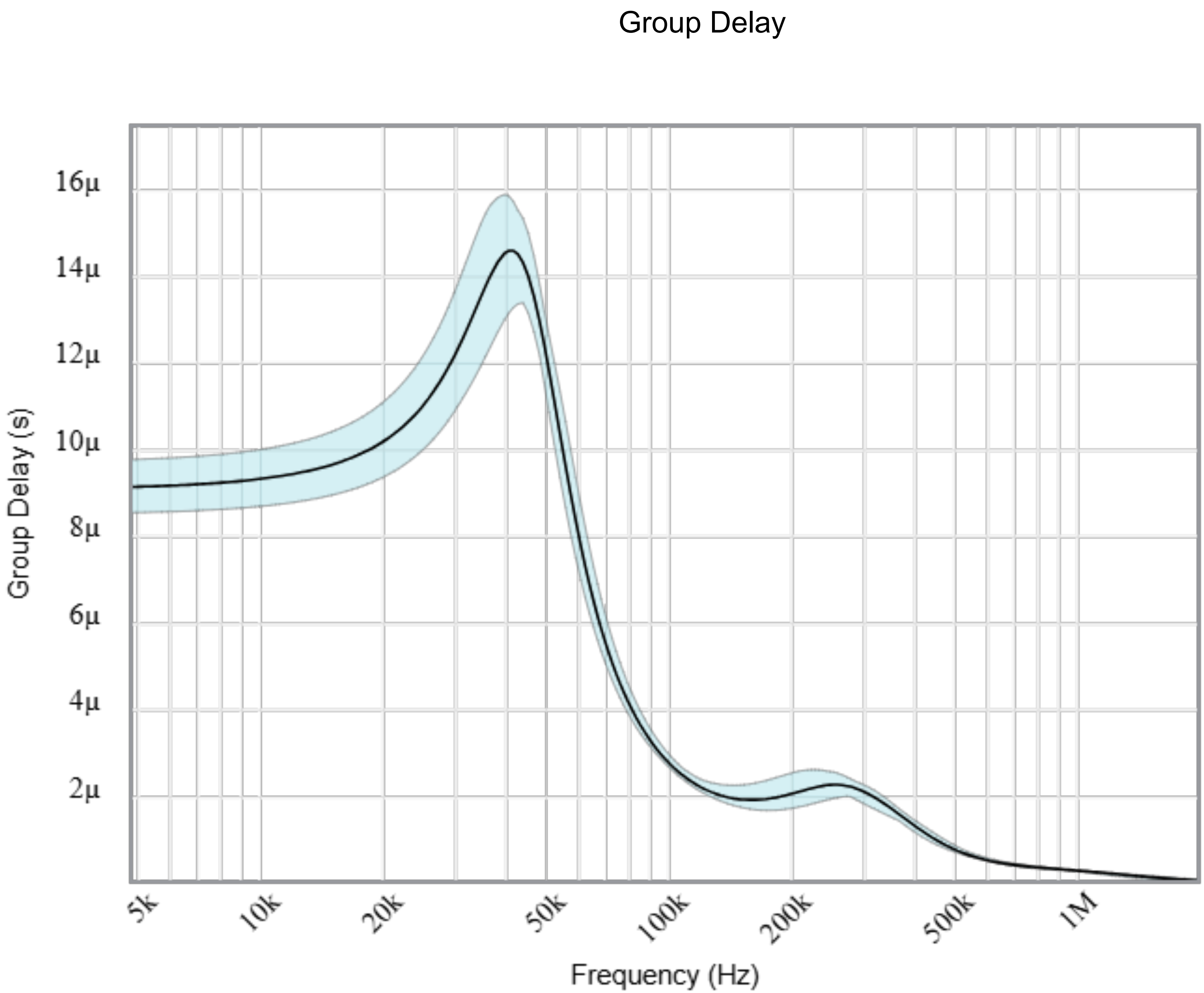


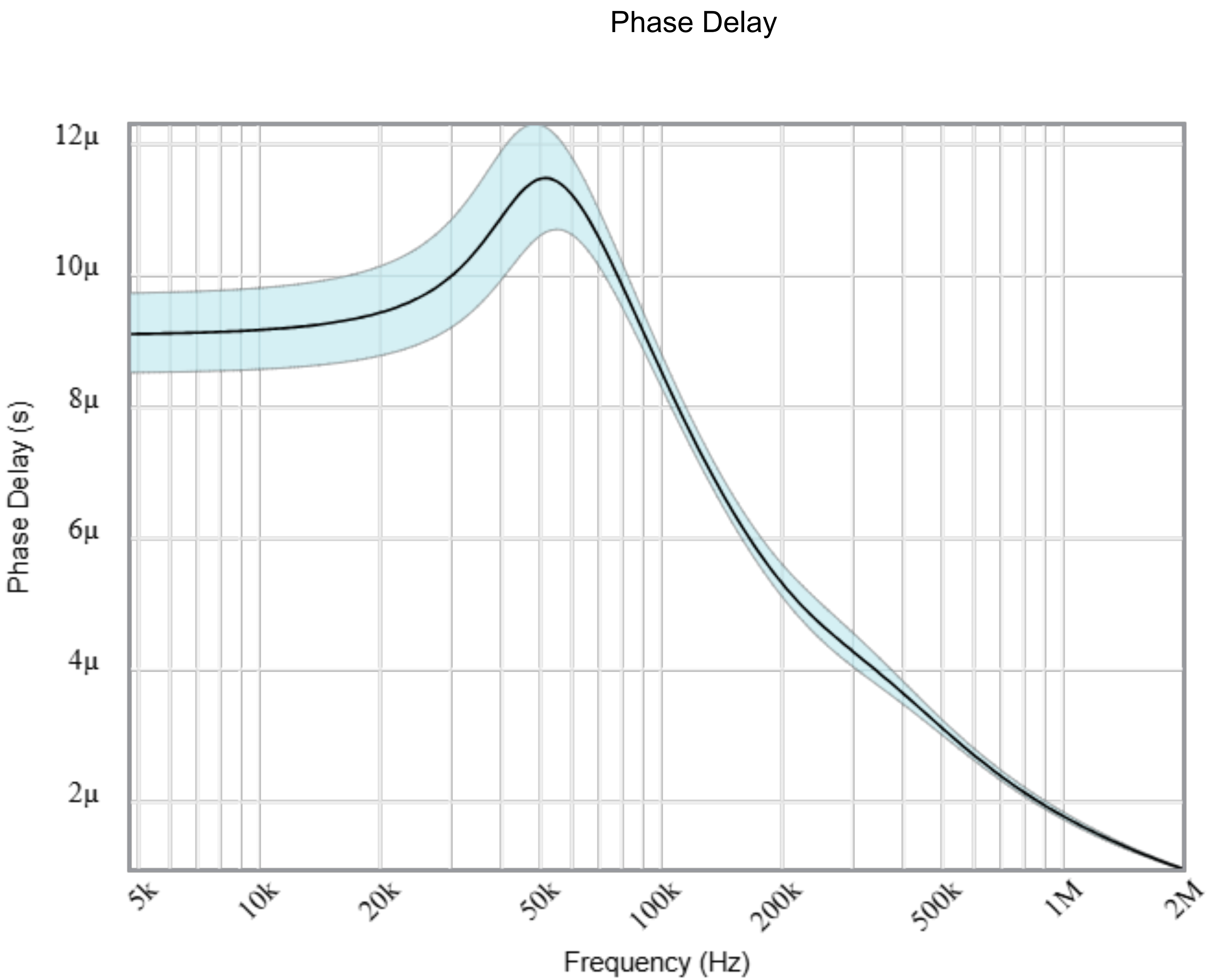
Phase(degrees)

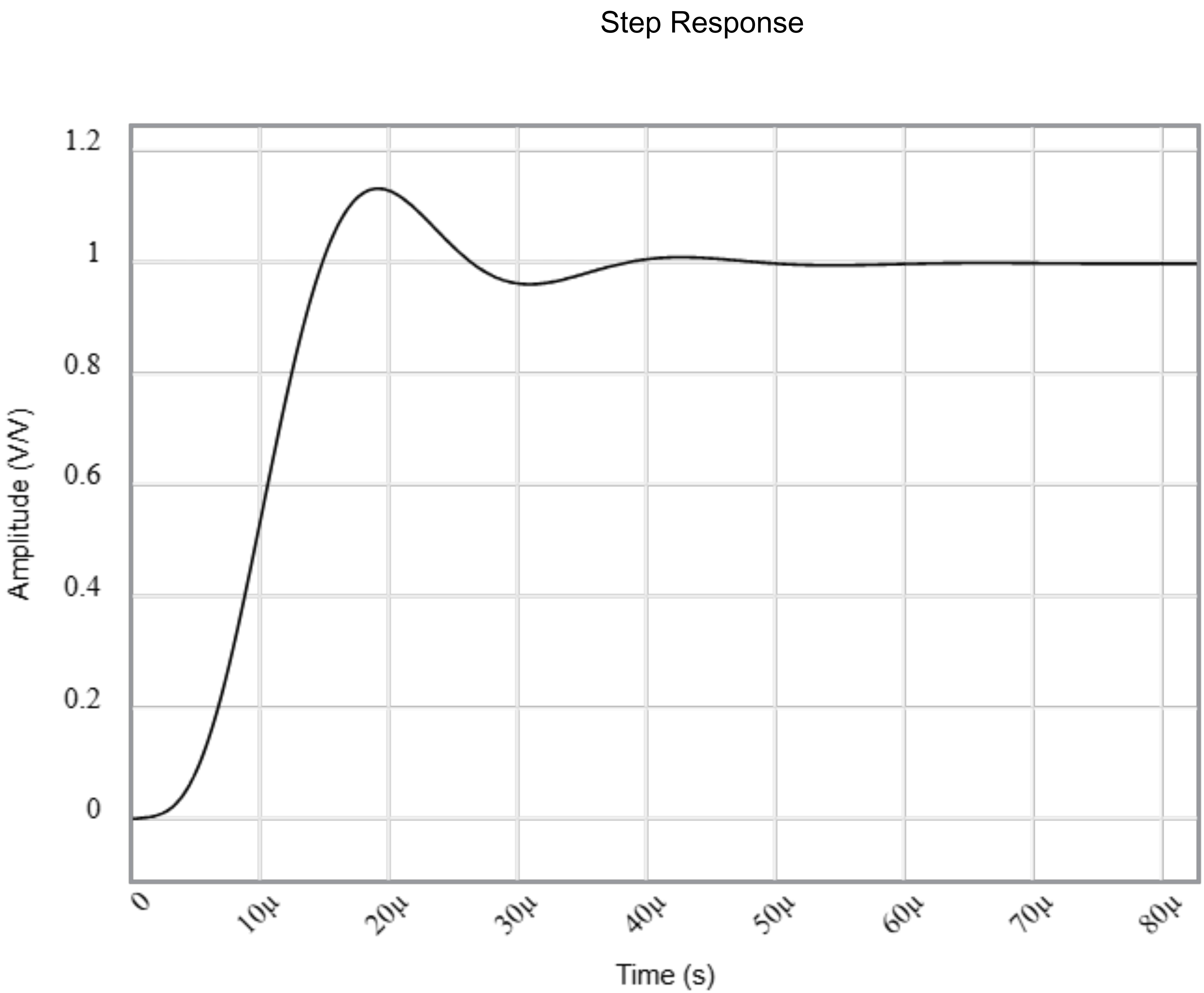


Phase(radians)

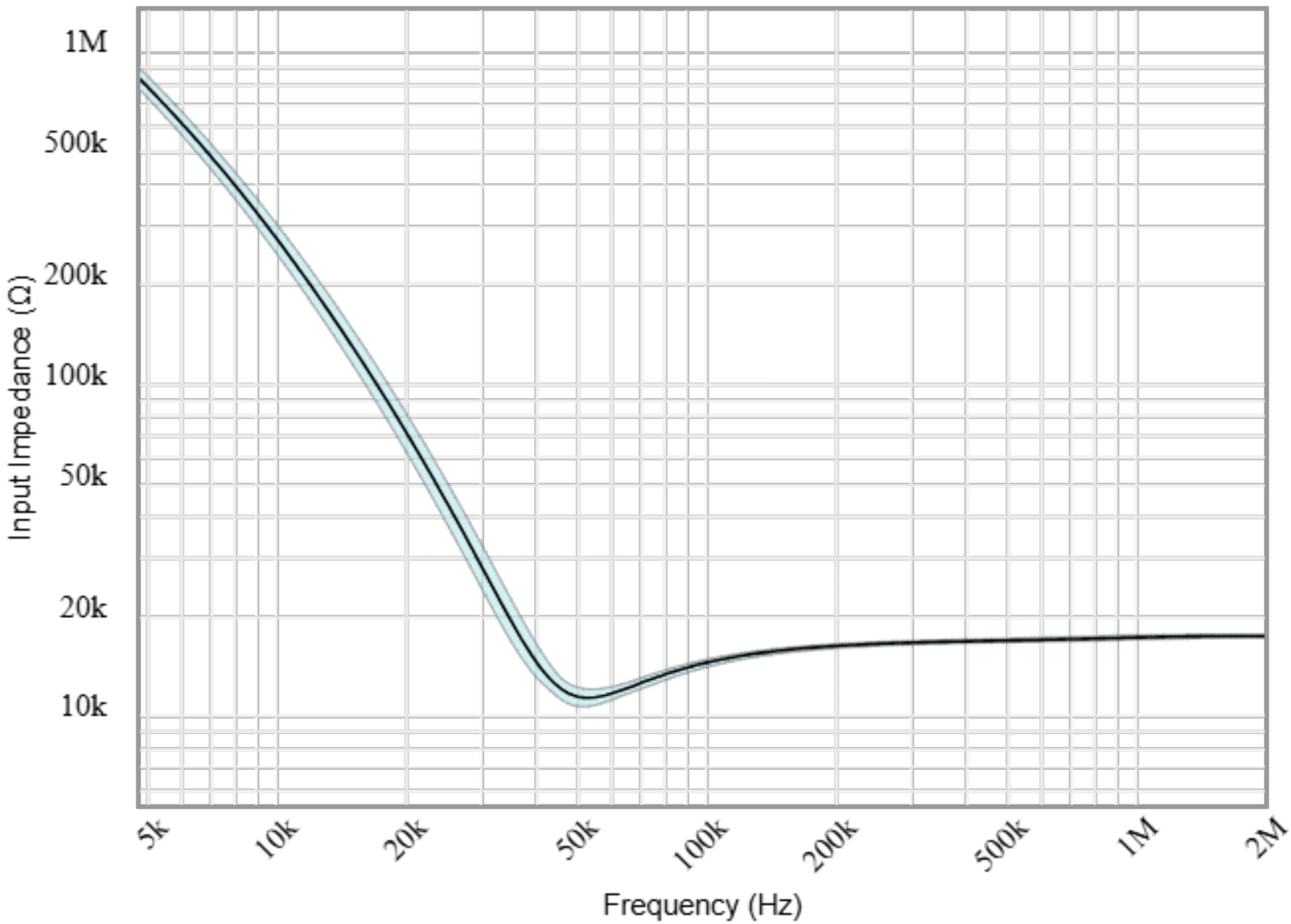


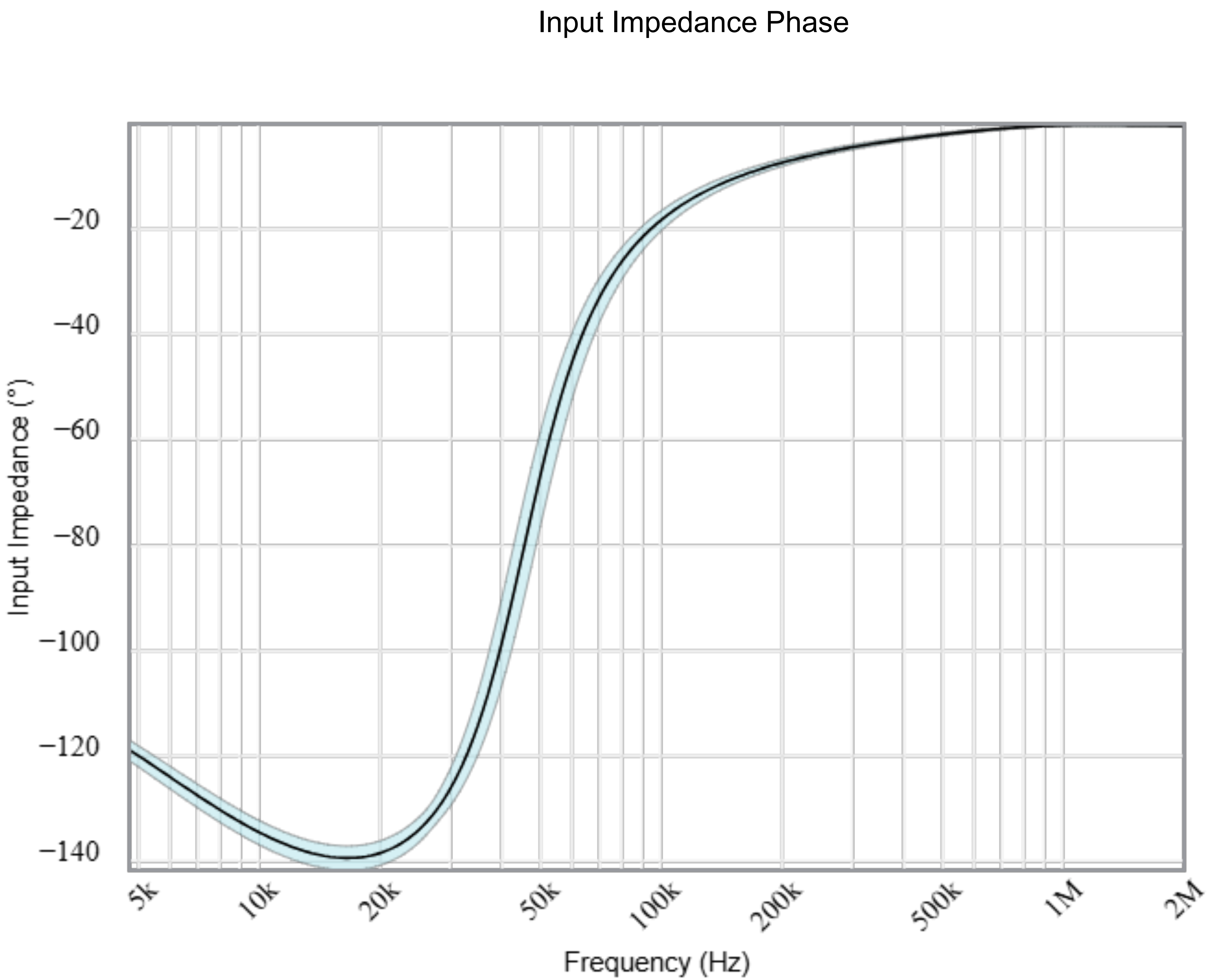

Group Delay
16µ
14µ
12µ
10µ
8µ
6µ
4µ
2µ
Group Delay (s)
5k
10k
20k
50k
100k
200k
500k
1M
Frequency (Hz)


Phase Delay
Phase Delay (s)
12µ
10µ
8µ
6µ
4µ
2µ
5k
10k
20k
50k
100k
200k
500k
1M
2M
Frequency (Hz)

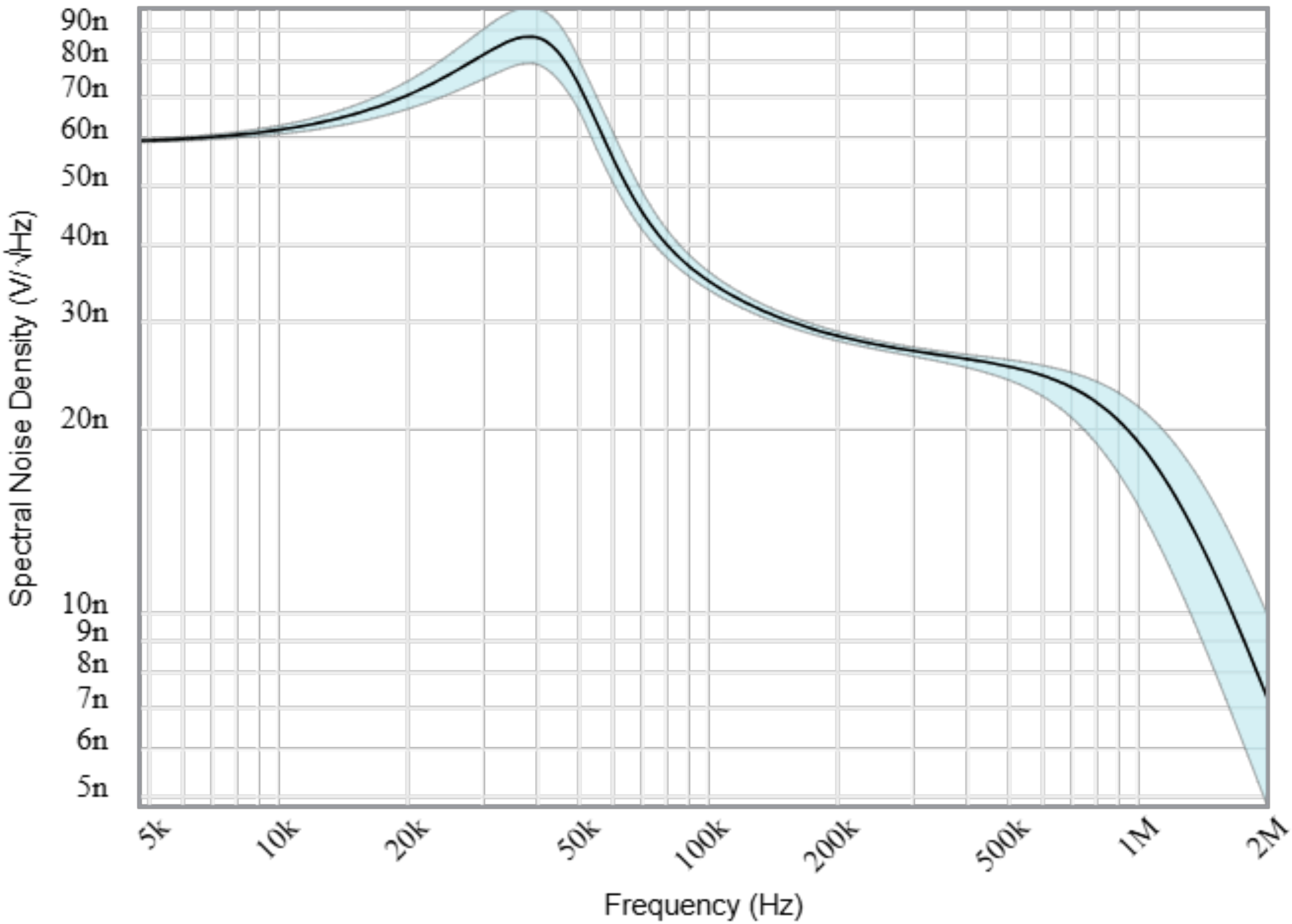

Step Response
Amplitude (V/V)
1.2
1
0.8
0.6
0.4
0.2
0
0
10µ
20µ
30µ
40µ
50µ
60µ
70µ
80µ
Time (s)

Input Impedance Magnitude




Input Impedance Phase
Input Impedance (°)
−20
−40
−60
−80
−100
−120
−140
5k
10k
20k
50k
100k
200k
500k
1M
2M
Frequency (Hz)

Noise

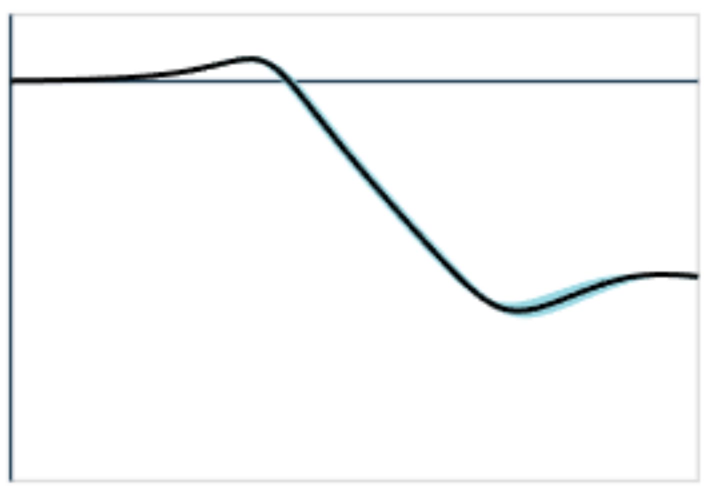

90n
80n
70n
60n
50n
40n
30n
20n
10n
9n
8n
7n
6n
5n
Spectral Noise Density (V/√Hz)
5k
10k
20k
50k
100k
200k
500k
1M
2M
Frequency (Hz)

# Stages

Your filter requires 2 op amp stage(s) with the following characteristics

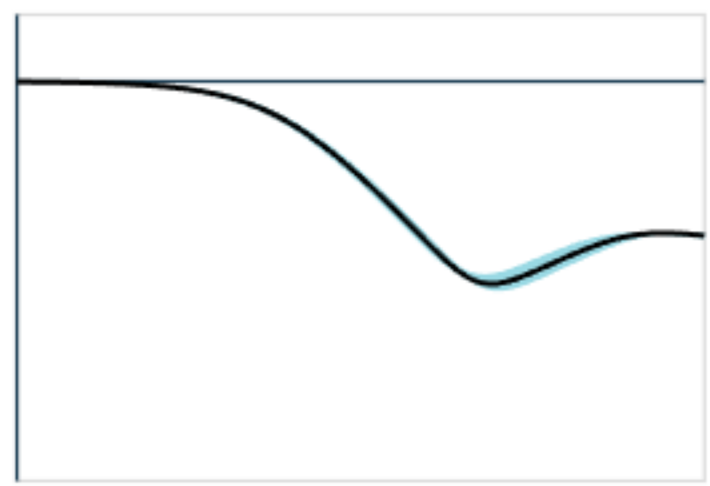
**A** — 2nd order Low-Pass Sallen Key

| | Target | Simulated |
|---|---|---|
| Gain (V/V): | 1 | 0.999 to 0.999 |
| $f_p$ (Hz): | 48k | 42.6k to 48.4k |
| Q: | 1.31 | 1.29 to 1.46 |

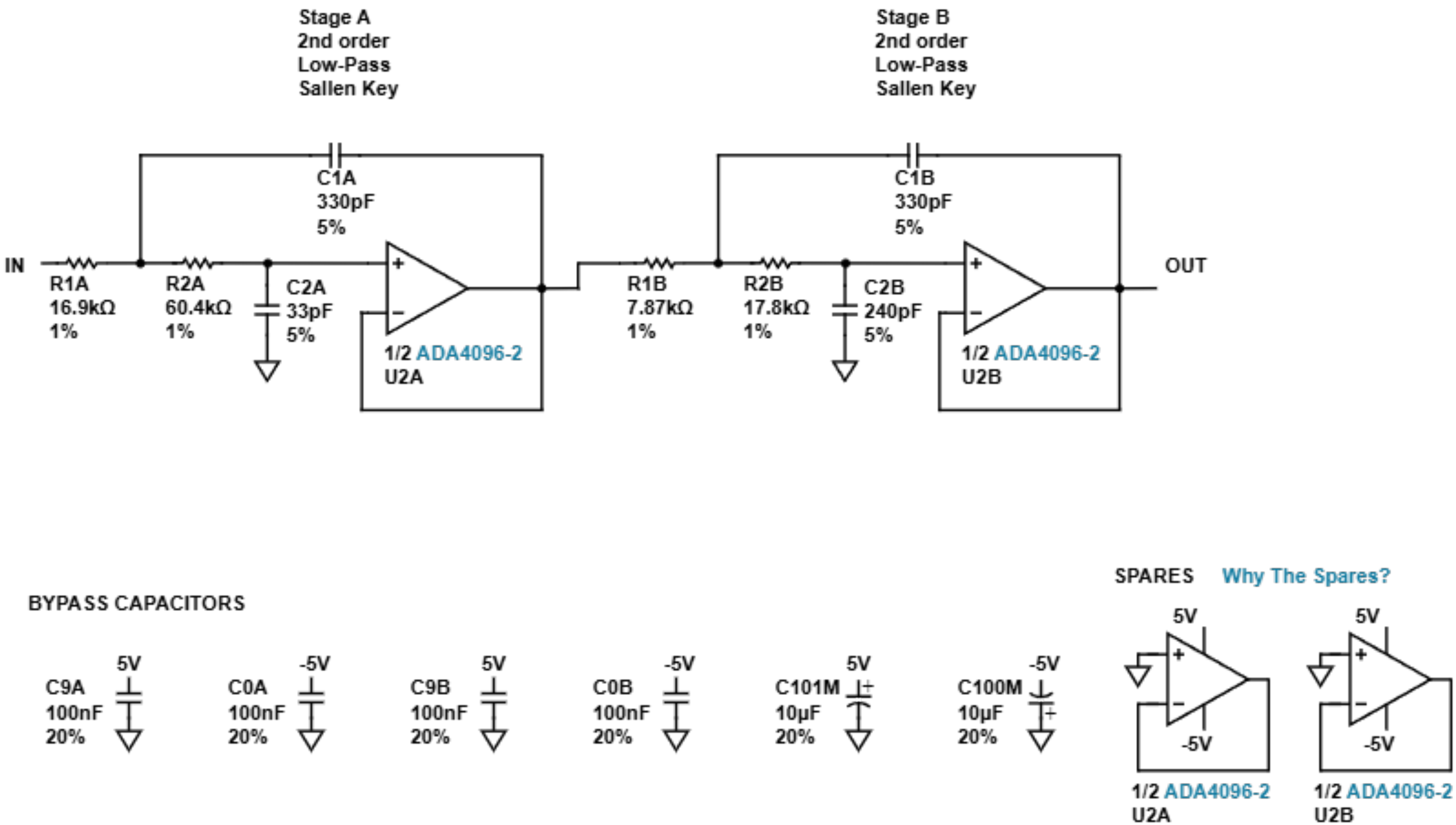


**B** — 2nd order Low-Pass Sallen Key

| | Target | Simulated |
|---|---|---|
| Gain (V/V): | 1 | 1 to 1 |
| $f_p$ (Hz): | 48k | 43.2k to 49.1k |
| Q: | 541m | 526m to 590m |



# Circuit


Stage A
2nd order
Low-Pass
Sallen Key
Stage B
2nd order
Low-Pass
Sallen Key
C1A
330pF
5%
C1B
330pF
5%
IN
R1A
16.9kΩ
1%
R2A
60.4kΩ
1%
C2A
33pF
5%
1/2 ADA4096-2
U2A
R1B
7.87kΩ
1%
R2B
17.8kΩ
1%
C2B
240pF
5%
1/2 ADA4096-2
U2B
OUT
BYPASS CAPACITORS
C9A
100nF
20%
5V
C0A
100nF
20%
-5V
C9B
100nF
20%
5V
C0B
100nF
20%
-5V
C101M
10µF
20%
5V
C100M
10µF
20%
-5V
SPARES
Why The Spares?
5V
-5V
1/2 ADA4096-2
U2A
5V
-5V
1/2 ADA4096-2
U2B